Sets, Divisibility Chains, and Closure Operators

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Several elementary set exercises in this note are really questions about order: what does a condition select, when is a family closed under an operation, and how do we construct the smallest closed object? We begin by checking the concrete exercises, then develop their common structure. The order-theoretic language follows Davey and Priestley, *Introduction to Lattices and Order*, particularly the chapters on closure operations, Galois connections, and fixed points.[1]

§1.1 Membership comes before manipulation

Let

$$A = \{0, 1, 2, 3\}, \qquad B = \{x \in A : 2 - x^2 \notin A\}.$$

There are exactly four candidates, all taken from A:

x	0	1	2	3
$2 - x^2$	2	1	-2	-7
$2 - x^2 \in A$	yes	yes	no	no

Thus $B = \{2, 3\}$. Checking negative inputs would be invalid because the defining condition already restricts $x \in A$.

More generally, for $f : X \to Y$, the image and preimage of a subset are

$$f(S) = \{f(s) : s \in S\}, \qquad f^{-1}(T) = \{x \in X : f(x) \in T\}.$$

They behave differently. For any $T_1, T_2 \subseteq Y$,

$$f^{-1}(T_1 \cap T_2) = f^{-1}(T_1) \cap f^{-1}(T_2),$$

whereas always

$$f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2).$$

The latter can be strict: with $f(x) = x^2$, $S_1 = \{-1\}$, $S_2 = \{1\}$, the left side is empty and the right side is $\{1\}$.

§1.2 Closure under the Chosen Operation

A subset $S \subseteq \mathbb{R}$ is closed under addition if $x, y \in S$ implies $x + y \in S$. Examples include $\mathbb{Z}, \mathbb{Q}, [0, \infty)$, and $(-\infty, 0]$. The last two are proper additive-closed subsets whose union is all of $\mathbb{R}$. Therefore the claim that two proper additive-closed sets can never cover $\mathbb{R}$ is false.

There is a related correct statement if the sets are *subgroups*. A group is not the union of two proper subgroups. To prove it, let H, K be proper subgroups. If $H \subseteq K$ or $K \subseteq H$, their union is plainly proper. Otherwise choose $h \in H \setminus K$ and $k \in K \setminus H$. Then

[1] Davey and Priestley, *Introduction to Lattices and Order*, chapter on complete lattices and Galois connections.

$h + k \notin H$, since otherwise $k = (h + k) - h \in H$; similarly $h + k \notin K$. This contradicts $H \cup K = G$.

The difference is structural: an additive submonoid need not contain additive inverses. Removing that hypothesis changes the conclusion.

§1.3 An extremal divisibility-chain exercise

Suppose $M \subseteq \{1, \dots, 2011\}$ is nonempty and for each $a, b \in M$ at least one of $a \mid b$, $b \mid a$, or $|a - b| = \max M$ holds. Write $m = \max M$. Since $1 \le a, b \le m$, one always has $|a - b| \le m - 1$, so the third option is impossible.

Thus all pairs in M must be comparable under divisibility. Enumerate its distinct elements in increasing order:

$$a_1 < a_2 < \cdots < a_r, \qquad a_i \mid a_{i+1}.$$

Every proper positive multiple is at least twice the original number; hence $a_{i+1} \ge 2a_i$ and inductively $a_r \ge 2^{r-1}$. Since $a_r \le 2011 < 2^{11}$, it follows that $r \le 11$. Equality occurs for

$$M = \{1, 2, 4, \dots, 1024\}.$$

Therefore the exact maximum cardinality is 11. The argument is an order-theoretic chain bound derived from the original arithmetic condition, without importing unrelated extremal theorems.

§1.4 Generated objects are intersections of closed objects

A closure operator $c : P \to P$ on a poset is extensive, monotone, and idempotent:

$$x \le c(x), \quad x \le y \Rightarrow c(x) \le c(y), \quad c(c(x)) = c(x).$$

For subsets of a vector space, $c(S) = \operatorname{span}(S)$ is one example; for a group, $c(S) = \langle S \rangle$ is another.

To see why intersections naturally arise, suppose $\mathcal{F}$ is a family of subsets closed under arbitrary intersections and containing the entire universe. Define

$$c(S) = \bigcap \{C \in \mathcal{F} : S \subseteq C\}.$$

The intersection is a member of $\mathcal{F}$ and contains S. Monotonicity follows because a larger input imposes a stricter condition on the indexing family; idempotence follows because $c(S)$ is already in $\mathcal{F}$. This gives an actual construction of the smallest closed object, rather than merely naming it.

§1.5 Image and preimage form an adjunction

The image/preimage relationship is more precise than a slogan about intersections. For $f : X \to Y$, $A \subseteq X$, and $B \subseteq Y$,

$$\boxed{f(A) \subseteq B \quad \Longleftrightarrow \quad A \subseteq f^{-1}(B).}$$

Both directions follow immediately from the definition of image and preimage. In the language of ordered sets, direct image is left adjoint to inverse image.

Consequently, the composite

$$c(A) = f^{-1}(f(A))$$

is a closure operator on the powerset of X: it includes A, is monotone, and is idempotent because $f(f^{-1}(f(A))) = f(A)$. What does it add? Exactly the points indistinguishable from points of A under f:

$$c(A) = \{x \in X : \exists a \in A,\ f(x) = f(a)\}.$$

For $f(x) = x^2$ and $A = \{2\}$ in $\mathbb{R}$, we get $c(A) = \{-2, 2\}$. This concrete calculation realizes an abstract Galois connection without invoking sophisticated machinery.

§1.6 Closure, fixed points, and stable iteration

A subset C is closed precisely when $c(C) = C$. The fixed points of a closure operator are closed under arbitrary intersections, because if $C_i = c(C_i)$ then monotonicity gives

$$c\left(\bigcap_i C_i\right) \subseteq \bigcap_i c(C_i) = \bigcap_i C_i,$$

and extensivity gives the reverse containment.

An apparently similar but distinct topic is fixed points of a general monotone map $F : \mathcal{P}(X) \to \mathcal{P}(X)$. There is a useful direct proof of existence of a *least* fixed point. Set

$$L = \bigcap \{A \subseteq X : F(A) \subseteq A\}.$$

The family is nonempty because X belongs to it. By monotonicity, $F(L) \subseteq L$. Applying F gives $F(F(L)) \subseteq F(L)$, so $F(L)$ itself belongs to the family defining L. Its minimality therefore implies $L \subseteq F(L)$. Hence $F(L) = L$. This is the least-fixed-point construction underlying the Knaster–Tarski theorem on complete lattices.

For a finite X, if the monotone process is also extensive and begins at S, the chain $S \subseteq F(S) \subseteq F^2(S) \subseteq \cdots$ must stabilize after finitely many strict increases. On an infinite universe, monotonicity alone does not warrant a claim of finite stabilization; a more careful limit construction may be required.

§1.7 A small algebraic side problem

The pair of elementary symmetric expressions

$$u = x + y, \qquad v = xy$$

determines the unordered pair $\{x, y\}$ as the roots of

$$t^2 - ut + v = 0.$$

Over $\mathbb{R}$, such a pair exists exactly when $u^2 - 4v \geq 0$. The discriminant is therefore a precise consistency condition on the proposed sums and products, rather than an unexplained substitution trick.

This is an example of the same general practice: specify which quantities are preserved under a transformation (here swapping x, y), then write the conditions in those invariants. For the set problems, the corresponding invariant information is closure under an operation or compatibility with image and preimage.